12.8 (a) Under appropriate conditions, we have

$$\int \vec{A}\, d^3x = -\int \vec{x}\,(\nabla\cdot\vec{A})\, d^3x \quad \text{for arbitrary vector function } \vec{A}(\vec{x}).$$

Then,

$$\vec{P}_{mech} = \int d^3x\, \gamma n m \vec{v} = -\int d^3x\, \vec{x}\,(\nabla\cdot(\gamma n m \vec{v})) = -\int d^3x\, \vec{x}\,(n\vec{v}\cdot\nabla(\gamma m))$$

From Eq. (12.2), we know

$$\frac{dE}{dt} = \frac{\partial E}{\partial t} + \vec{v}\cdot\nabla E = \vec{v}\cdot\nabla E = e\vec{v}\cdot\vec{E},$$

the mechanical momentum becomes, noticing that $E = \gamma m c^2$

$$\vec{P}_{mech} = -\frac{1}{c^2}\int d^3x\, \vec{x}\,(ne\vec{v}\cdot\vec{E}) = \frac{1}{c^2}\int d^3x\, \vec{x}\,(ne\vec{v}\cdot\nabla\Phi)$$

Using integration by parts, or the inverse. we have

$$\vec{P}_{mech} = -\frac{1}{c^2}\int d^3x\, ne\vec{v}\,\Phi = -\frac{1}{c^2}\int d^3x\, \Phi\vec{J}.$$